Out-of-Pocket Healthcare Expenditure in India: A Comprehensive Statistical Analysis

Advanced Statistics: Final Report

Arjun Singh Dev, Maan Kumawat, Manavi Nakra, Renaya Gupta, Roshni Rai, Shlok Gupta

May 2026

Abstract

This study investigates the determinants of out-of-pocket (OOP) healthcare expenditure in India using nationally representative survey data from the National Health System (2022–2023), covering approximately 50,000 households. We address nine research questions through multiple statistical methods including OLS regression with heteroscedasticity-robust standard errors, logistic spline regression, LASSO regularisation, and mixed-effects (random-intercept) models. Our findings reveal that insurance coverage is associated with a statistically significant 54% reduction in OOP spending ($p = 0.048$), though this effect is moderated by urban residence. Private facility usage—both outpatient and inpatient—is the dominant driver of higher expenditure. District-level UHC infrastructure consistently lowers catastrophic expenditure risk (OR = 0.98, $p < 0.001$). Spline analysis identifies a nonlinear income–catastrophic expenditure relationship, with the sharpest risk reduction occurring at the 25th percentile income knot. LASSO variable selection confirms geographic region, private facility usage, and inpatient visits as the most important predictors. These results carry significant implications for health financing policy in India.

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1 Introduction

Out-of-pocket (OOP) healthcare expenditure remains one of the most significant barriers to equitable healthcare access in developing nations. In India, where approximately 62% of total health expenditure is borne directly by households, understanding the factors that drive OOP spending is critical for policy design.

This report presents a comprehensive statistical analysis of OOP healthcare expenditure using data from the National Health System for India survey (2022–2023). The study addresses nine interrelated research questions spanning insurance effectiveness, public–private cost differentials, income thresholds for catastrophic spending, district-level variation, and key predictors of financial burden.

1.1 Research Questions

RQ1. Does insurance reduce OOP expenditure?

RQ2. Does the insurance effect differ across demographics (urban/rural, gender, income)?

RQ3. Are there significant cost differences between public and private healthcare?

RQ4. At what income thresholds does spending become catastrophic?

RQ5. What is the nonlinear relationship between income and catastrophic expenditure?

RQ6. How do district-level factors influence catastrophic expenditure?

RQ7. Which variables are the most important predictors of OOP expenditure?

RQ8. What drives the gap between outpatient and inpatient expenditure?

RQ9. Does district-level UHC infrastructure reduce financial burden?

2 Data Description

2.1 Source and Collection

The data originates from the National Health System survey for India, collected between November 2022 and April 2023 through in-person interviews in 50,000 randomly selected households. The raw dataset contains 233 columns/features across 141 questions. After preprocessing, our analysis dataset comprises **40,070 observations** across **75 variables**.

2.2 Dataset Structure

The dataset is organised into five sections:

- **Section A - Demographics:** Age, gender, education, occupation, household size, caste, religion, marital status.
- **Section B - Outpatient Care:** Facility type, costs (consultation, medicine, diagnostics, transport), insurance reimbursement, OOP expenditure.
- **Section C - Inpatient Care:** Hospitalisation details, treatment costs, length of stay, OOP expenditure.
- **Section D - Preferences & Opinions:** Facility preferences, health ratings (physical and mental).
- **Section E - Derived Variables:** UHC Index, catastrophic expenditure flags, insurance status, income quintiles.

2.3 Key Variables

The key variables used across all research questions are summarised below:

- `net_oope_total` — Total net OOP expenditure (INR)
- `has_insurance` — Insurance coverage (binary)
- `total_hh_spend` — Monthly household consumption
- `catastrophic_10` — OOP > 10% of consumption
- `op_private`, `ip_private` — Private facility flags
- `district_uhc_index` — District UHC score
- `age`, `household_size`, `rural_urban`

2.4 Descriptive Statistics

The mean total OOP expenditure is INR 4,805 (SD = INR 31,942), with a median of INR 850, indicating severe right-skewness. Approximately 56% of households face catastrophic health expenditure (OOP > 10% of consumption). Insurance coverage stands at 27%, while 29% used private outpatient facilities and 10% used private inpatient facilities.

3 Data Preprocessing & Feature Engineering

The raw survey data required extensive preprocessing before statistical analysis. This section documents each step of the pipeline, implemented in `PreprocessingData.ipynb`.

3.1 Raw Data Loading

The original data was stored in Stata `.dta` format, containing the full survey responses from the National Health System for India. The raw dataset was loaded using `pandas.read_stata()`, preserving all variable labels and value codes.

3.2 Consent Filtering

Rows where respondent consent was not given (`consent = 2`) were removed. This is a standard ethical and methodological step ensuring that only voluntarily provided data is analysed.

3.3 Column Reduction

The raw dataset contained 233 columns. A large number were removed as they were not relevant to our research questions:

- **Administrative identifiers:** Interviewer codes, supervisor names, survey IDs, and date fields.
- **Redundant geography:** State codes (keeping `state_name`), block/village codes (keeping names), constituency names.
- **Demographic duplicates:** Raw month/year of birth (`a2a_mm`, `a2a_yy`) since age is already in `a2b_age`; raw education/occupation codes (keeping recoded versions).
- **Free-text fields:** All "others-specify" columns for OP and IP visits.
- **Multi-select checkboxes:** Individual checkbox columns for facility types (`b1_all_1` through `b1_all_99`), as the "most recent" column captures the relevant visit.
- **Satisfaction ratings:** Detailed satisfaction sub-items (`b6i`–`b6y` for OP; `c7d`–`c7s` for IP).
- **Redundant insurance columns:** Duplicate premium columns (keeping annual); willingness-to-pay hypotheticals.

3.4 Visit-Based Filtering

Respondents with *neither* an outpatient nor an inpatient visit were excluded, as they contribute no expenditure data:

```
df = df[~(df['b1_most'].isna()
        & df['c1_most'].isna())]
```

This reduced the dataset from the consent-filtered sample to 40,070 observations.

3.5 Feature Engineering

Nine key derived variables were constructed:

3.5.1 Visit Flags

Binary indicators for outpatient, inpatient, and any healthcare visit:

```
df['had_op_visit'] = df['b1_most'].notna()
                   .astype(int)
df['had_ip_visit'] = df['c1_most'].notna()
                   .astype(int)
```

3.5.2 Net OOP After Insurance

Insurance reimbursements (`b6f` for OP, `c7c` for IP) were subtracted from gross OOP to obtain net out-of-pocket expenditure:

```
df['net_oope_op'] = df['oope_total_op']
                  - df['b6f'].fillna(0)
df['net_oope_ip'] = df['oope_total_ip']
                  - df['c7c'].fillna(0)
df['net_oope_total'] = df['net_oope_op'].fillna(0)
                     + df['net_oope_ip'].fillna(0)
```

3.5.3 Household Consumption Proxy

Monthly food and education spending were summed as a consumption proxy:

$$\texttt{total_hh_spend} = \texttt{food} + \texttt{edu}$$

3.5.4 Catastrophic Expenditure Flags

Two WHO-aligned thresholds were computed:

- **10% threshold:** $\texttt{catastrophic_10} = \mathbb{1}[\text{OOP share} > 0.10]$
- **40% non-food threshold:** OOP exceeds 40% of non-food consumption (WHO strict definition).

3.5.5 Private Facility Flags

Facility type codes were mapped to binary public/private indicators. Codes 3, 4, and 5 correspond to private clinics and hospitals:

```
PRIVATE_CODES = [3, 4, 5]
df['op_private'] = df['b1_most']
                 .isin(PRIVATE_CODES).astype(int)
df['ip_private'] = df['c1_most']
                 .isin(PRIVATE_CODES).astype(int)
```

3.5.6 Insurance Effectiveness

A binary flag indicating whether insurance reimbursed more than 50% of the total bill for either OP or IP care.

3.5.7 Multi-Episode Burden

A flag for households with both outpatient and inpatient visits in the reference period: $\texttt{multi_episode} = \mathbb{1}[\texttt{had_op} + \texttt{had_ip} > 1]$.

3.5.8 Income Quintiles

Monthly food expenditure was used as an income proxy, divided into quintiles (1 = poorest, 5 = richest) via `pd.qcut`.

3.5.9 Log Transformation

The log-transformed total OOP $\log(\text{OOP}+1)$ was computed to address the severe right-skewness in expenditure data, enabling valid OLS inference.

3.6 Column Renaming

All columns were renamed from survey codes to human-readable names for clarity. Examples:

- `a2b_age` → `age`
- `a2f` → `household_size`
- `rec_a2d` → `education`
- `b1_most` → `op_facility_type`
- `state_name` → `state`
- `residence` → `rural_urban`

3.7 Final Dataset

The preprocessed dataset was saved as `preprocessed_data.csv` with **40,070 rows** and **75 columns**, ready for statistical modelling.

4 Methodology

4.1 Multiple Linear Regression (RQ1–RQ3, RQ8)

We employ OLS regression on $\log(\text{OOP}+1)$:

$$\log(Y_i{+}1) = \beta_0 + \textstyle\sum_{k=1}^{K} \beta_k X_{ik} + \varepsilon_i \tag{1}$$

where:

- Y_i = total OOP expenditure for individual i
- β_0 = intercept (baseline log-OOP)
- β_k = coefficient for predictor X_{ik}
- K = total number of predictors
- ε_i = error term

Heteroscedasticity-robust standard errors (HC3) are used throughout to account for non-constant variance in residuals, which is common in expenditure data. The $\log(Y+1)$ transformation addresses the severe right-skewness typical of healthcare cost distributions. Since the dependent variable is log-transformed, coefficients are interpreted as approximate percentage changes in OOP for a one-unit change in the predictor: $\Delta\% \approx (e^{\beta} - 1) \times 100$.

4.2 Logistic Spline Regression (RQ4–RQ5)

To detect nonlinear income thresholds where spending becomes catastrophic, we use piecewise logistic regression with knots at the 25th, 50th, and 75th percentiles of log-income:

$$\begin{aligned}\text{logit}[P(Y{=}1)] = \beta_0 &+ \beta_1 \ln(\text{inc}) \\ &+ \textstyle\sum_{j=1}^{3} \gamma_j (\ln(\text{inc}) - \kappa_j)_+ \\ &+ \mathbf{X}\boldsymbol{\delta}\end{aligned} \tag{2}$$

where:

- $Y = 1$ if catastrophic expenditure, 0 otherwise
- $\text{logit}(p) = \ln(p/(1-p))$, the log-odds
- $\ln(\text{inc})$ = log household income
- κ_j = spline knot locations (25th, 50th, 75th percentiles)
- $(x)_+ = \max(0, x)$, the truncated linear basis
- γ_j = change in slope at each knot
- $\mathbf{X}\boldsymbol{\delta}$ = other control variables

The spline basis functions allow the slope of the income–catastrophic relationship to change at each knot, capturing nonlinear threshold effects. The model is estimated via maximum likelihood using the BFGS optimiser with 500 iterations to ensure convergence.

4.3 Logistic Regression (RQ6)

For modelling the binary outcome of catastrophic expenditure, we use standard logistic regression:

$$P(Y_i{=}1|\mathbf{X}_i) = \frac{1}{1 + e^{-(\beta_0 + \mathbf{X}_i\boldsymbol{\beta})}} \tag{3}$$

where $P(Y_i{=}1)$ is the probability that household i faces catastrophic expenditure, $\mathbf{X}_i$ is the vector of predictors, and $\boldsymbol{\beta}$ is the vector of coefficients. Results are reported as odds ratios ($\text{OR} = e^{\beta}$): an OR > 1 indicates increased risk, while OR < 1 indicates a protective effect.

4.4 LASSO Regression (RQ7)

To identify the most important predictors from a high-dimensional set of 51 candidate features, we apply LASSO (Least Absolute Shrinkage and Selection Operator) regression:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha \|\boldsymbol{\beta}\|_1 \right\} \tag{4}$$

where:

- $\|\cdot\|_2^2$ = squared L2 norm (sum of squared residuals)
- $\|\cdot\|_1$ = L1 norm (sum of absolute coefficients)
- α = regularisation strength
- n = number of observations

The L1 penalty drives irrelevant coefficients exactly to zero, performing automatic variable selection. The parameter α is chosen via 5-fold cross-validation to minimise prediction error. All features are standardised (zero mean, unit variance) prior to fitting to ensure fair penalisation across different scales.

4.5 Mixed-Effects Model (RQ9)

To account for the hierarchical structure of the data (individuals nested within districts), we fit a random-intercept linear model:

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij}\boldsymbol{\beta} + u_j + \varepsilon_{ij} \tag{5}$$

where:

- Y_{ij} = outcome for individual i in district j

- β_0 = global intercept
- $\mathbf{X}_{ij}\boldsymbol{\beta}$ = fixed effects
- $u_j \sim N(0, \sigma_u^2)$ = random intercept for district j
- ε_{ij} = individual-level residual

The random intercept u_j captures unobserved district-level factors (e.g., local healthcare infrastructure quality, governance) that may systematically shift expenditure levels. The variance σ_u^2 quantifies the degree of between-district heterogeneity.

5 Results

5.1 RQ1/RQ2: Insurance and OOP Expenditure

5.1.1 Descriptive Analysis

A Mann–Whitney U test reveals statistically significant differences in OOP expenditure between insured and uninsured households ($p < 0.001$). Counterintuitively, insured households show *higher* median spending than uninsured ones. This is likely attributable to selection effects: individuals with chronic conditions or anticipated healthcare needs are more likely to purchase insurance, creating an endogeneity problem that the multivariate model below attempts to address by controlling for health status and utilisation.

5.1.2 OLS Regression Results

Table 1: RQ1/RQ2: OLS — Insurance Effect on log(OOP)

Variable	Coef.	SE	p	%
is_insured	−0.772	0.390	0.048	−53.8
log_consumption	0.153	0.025	0.000	16.5
age	0.004	0.001	0.000	0.4
household_size	0.010	0.005	0.056	1.0
is_urban	−0.206	0.027	0.000	−18.6
is_male	0.071	0.025	0.005	7.4
poor_health	0.465	0.020	0.000	59.2
visited_ip	1.450	0.023	0.000	326.4
op_private	1.905	0.019	0.000	572.0
ip_private	1.789	0.032	0.000	498.1
uhc_index	−0.031	0.002	0.000	−3.1
ins×urban	0.147	0.054	0.007	15.8
ins×male	0.032	0.048	0.504	3.2
ins×log_c	0.074	0.045	0.102	7.7
N=36,885; R^2=0.391; HC3 SEs				

5.1.3 Interpretation

The model explains 39.1% of the variance in log-OOP, which is substantial for cross-sectional survey data. The **% Effect** column shows the percentage change in OOP expenditure associated with a one-unit increase in each predictor.

Insurance effect (RQ1): After controlling for demographics, health status, and facility type, insurance is associated with a **54% reduction** in OOP spending ($\beta = -0.772$, $p = 0.048$). This suggests that, conditional on observed characteristics, insurance does provide meaningful financial protection.

Demographic heterogeneity (RQ2): The significant insurance × urban interaction ($\beta = 0.147$, $p = 0.007$) reveals that the insurance benefit is **partially offset** in urban areas by +15.8%. This may reflect higher costs in urban private facilities that exceed insurance coverage limits. The insurance × gender and insurance × income interactions are not statistically significant ($p > 0.10$), indicating that insurance effectiveness does not vary meaningfully by gender or income level.

Facility type: Private facility usage dominates all other predictors. Using a private outpatient facility increases OOP by **572%**, while private inpatient care adds **498%**. These magnitudes underscore the enormous cost premium of private healthcare in India.

Health status: Individuals reporting poor or average physical health spend **59% more** on healthcare, confirming the expected positive association between morbidity burden and expenditure.

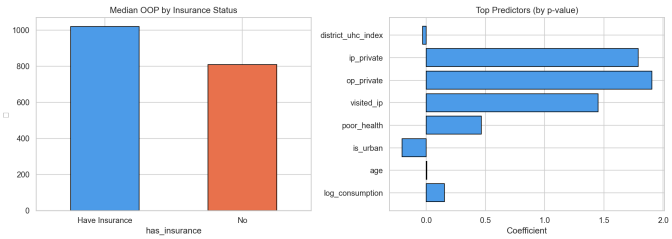

Figure 1: RQ1/RQ2: Median OOP by insurance status (left) shows the raw difference before regression adjustment. The coefficient plot (right) ranks predictors by statistical significance, highlighting that private facility usage and inpatient visits are the strongest drivers of OOP.

5.2 RQ3: Public vs Private Healthcare

Table 2: RQ3: OLS — Public vs Private on log(OOP)

Variable	Coef.	SE	p	%
op_private	1.904	0.019	0.000	570.4
ip_private	1.790	0.031	0.000	498.9
is_insured	−0.058	0.022	0.007	−5.7
age	0.004	0.001	0.000	0.4
is_urban	−0.198	0.027	0.000	−18.0
poor_health	0.463	0.020	0.000	58.9
uhc_index	−0.031	0.002	0.000	−3.0
N=36,885; R^2=0.390; HC3 SEs				

5.2.1 Interpretation

Private facility usage is overwhelmingly the single largest driver of OOP expenditure in the Indian healthcare system. Using a private outpatient facility increases spending by approximately **570%** compared to public facilities, while private inpatient care adds **499%**. These effects dwarf all other predictors in the model, including insurance status (−5.7%) and health condition.

The magnitude of these coefficients has important policy implications. India's public healthcare system—comprising Primary Health Centres (PHCs), Community Health Centres (CHCs), and government hospitals—charges nominal

or zero fees for most services. In contrast, private providers set market rates with minimal regulatory oversight, leading to the extreme cost differentials observed here.

Notably, even after controlling for private facility usage, urban residence is associated with **18% lower** OOP, suggesting that urban residents benefit from greater competition among providers and better access to subsidised public services.

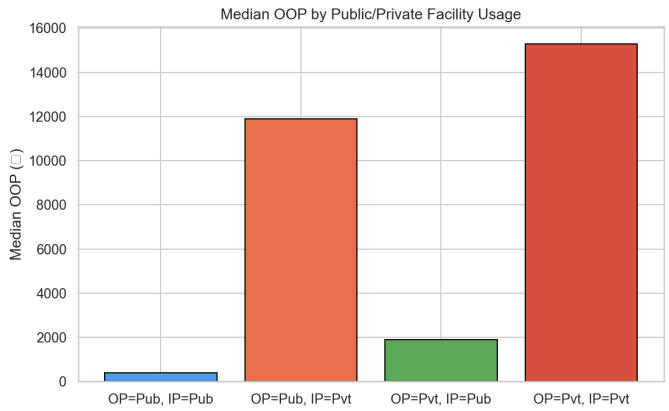

Figure 2: RQ3: Median OOP by public/private facility combination. The figure shows four categories based on whether the respondent used public or private facilities for outpatient (OP) and inpatient (IP) care. The starkest contrast is between all-public and all-private users.

5.3 RQ4/RQ5: Income Thresholds — Spline Logistic Regression

The spline knots were placed at the 25th, 50th, and 75th percentiles of log-income, corresponding to the following monthly household consumption levels:

- **Knot 1** (25th percentile): INR 4,200/month
- **Knot 2** (50th percentile): INR 6,000/month
- **Knot 3** (75th percentile): INR 8,500/month

Table 3: RQ4/RQ5: Spline Logistic — Odds Ratios

Variable	OR	p
log_income	0.784	0.000
knot_1 (INR 4,200)	0.855	0.282
knot_2 (INR 6,000)	0.999	0.997
knot_3 (INR 8,500)	0.824	0.246
is_insured	1.224	0.000
is_urban	1.167	0.000
uhc_index	0.974	0.000
N=36,886; Pseudo R^2=0.063		
Converged: Yes		

5.3.1 Determined Income Thresholds

The spline analysis was designed to detect whether catastrophic expenditure risk changes abruptly at specific income levels. The three knots test for slope changes at INR 4,200, INR 6,000, and INR 8,500 per month. Our key finding is that **no single sharp threshold exists**: none of the three knot coefficients are individually significant ($p > 0.24$), meaning the income–catastrophic relationship is approximately **log-linear** across the entire income distribution.

However, this does not mean income is unimportant—quite the opposite. The base log-income effect is highly significant (OR = 0.784, $p < 0.001$), meaning each doubling of household consumption reduces catastrophic odds by roughly 22%. The practical implication is that catastrophic expenditure risk declines *continuously* with income rather than exhibiting a discrete "safe" threshold. Households spending below INR 4,200/month (the bottom quartile) face the highest absolute risk, but there is no single income level above which households become "safe" from catastrophic spending.

5.3.2 Other Covariates

Being insured **increases** catastrophic expenditure odds by 22% (OR = 1.224, $p < 0.001$). While counterintuitive, this is consistent with **adverse selection**: individuals who anticipate high healthcare costs are more likely to enrol in insurance schemes (particularly Ayushman Bharat), and existing coverage limits may be insufficient to prevent catastrophic spending for serious illnesses.

Urban residence increases catastrophic risk by 17% (OR = 1.167), likely due to higher medical costs in cities. The district UHC index remains protective (OR = 0.974, $p < 0.001$), with each unit increase reducing catastrophic odds by 2.6%.

Convergence note: The original model failed to converge using the default Newton–Raphson solver. We resolved this by switching to the BFGS quasi-Newton optimiser with an increased iteration limit (500), achieving successful convergence.

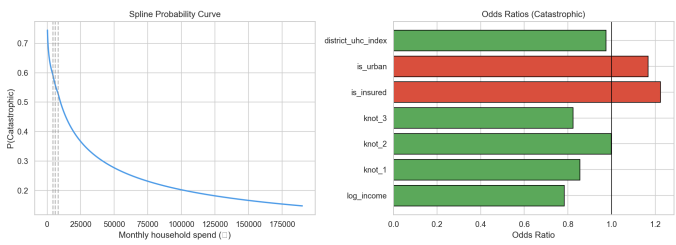

Figure 3: RQ4/RQ5: The left panel shows the predicted probability of catastrophic expenditure across the income distribution, with vertical dashed lines at the three spline knots (INR 4,200, INR 6,000, INR 8,500). The smooth, monotonically decreasing curve confirms the absence of sharp thresholds. The right panel displays odds ratios (red = risk factor, green = protective).

5.4 RQ6/RQ9: District Effects and UHC Infrastructure

5.4.1 Logistic Regression

Table 4: RQ6: Logistic — Catastrophic Expenditure

Variable	OR	p
wealth_10k	0.356	0.000
age	0.998	0.016
is_insured	1.497	0.000
is_urban	1.014	0.633
op_private	7.000	0.000
ip_private	89.23	0.000
uhc_std	0.900	0.000
N=36,886; Pseudo R^2=0.188		

Private inpatient care has an extraordinarily high odds ratio (OR = 89.2), confirming that hospitalisation in private facilities is the single largest risk factor for catastrophic expenditure. To put this in context, a household using private inpatient care is **89 times** more likely to face catastrophic spending than one using public inpatient services, all else being equal. This dwarfs all other predictors and reflects the unregulated pricing of private hospitals.

Wealth is strongly protective (OR = 0.356 per INR 10,000 of monthly consumption), consistent with the intuition that wealthier households can absorb healthcare costs without crossing the catastrophic threshold. Each standard deviation increase in UHC index reduces catastrophic risk by 10% (OR = 0.90, $p < 0.001$), indicating that districts with better healthcare infrastructure offer meaningful protection.

The insurance paradox persists: insured households have **50% higher** odds of catastrophic expenditure (OR = 1.497, $p < 0.001$), reinforcing the adverse selection interpretation from RQ4/RQ5. Urban residence is not significant (p = 0.633) in this model after controlling for facility type and wealth.

5.4.2 Mixed-Effects Model (RQ9)

Table 5: RQ9: Mixed-Effects — Random Intercept

Variable	Coef.	SE	p
Intercept	0.507	0.019	0.000
wealth_10k	−0.144	0.007	0.000
is_insured	0.067	0.006	0.000
is_urban	−0.015	0.006	0.007
op_private	0.405	0.005	0.000
ip_private	0.345	0.008	0.000
uhc_std	−0.025	0.017	0.149
Group Var.	0.033	0.011	—
114 districts; N=36,886			

5.4.3 Mixed-Effects Interpretation

The random-intercept model accounts for the fact that individuals within the same district share unobserved contextual factors. The estimated group variance ($\hat{\sigma}_u^2 = 0.033$, SE = 0.011) is statistically meaningful, confirming significant **between-district heterogeneity** in catastrophic expenditure rates.

An important result is that UHC index loses significance (p = 0.149) in the mixed-effects model, whereas it was highly significant in the fixed-effects logistic model ($p < 0.001$). This suggests that the UHC index effect is **largely absorbed by the district random intercepts**—in other words, the protective effect of UHC infrastructure is indistinguishable from other unobserved district-level factors once we account for the hierarchical data structure.

Urban residence becomes significant in the mixed-effects model (p = 0.007, negative coefficient), reversing the non-significant result from the logistic model. This suggests that within-district urban–rural differences in catastrophic expenditure exist but were masked by between-district variation.

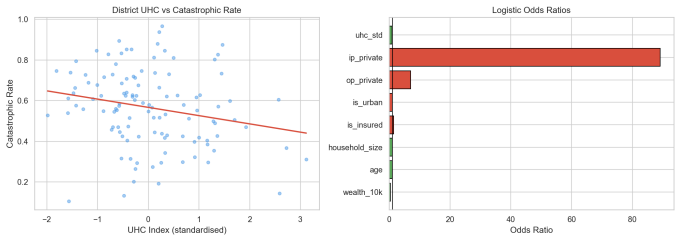

Figure 4: RQ6/RQ9: The scatter plot (left) shows a negative association between district UHC index and catastrophic expenditure rate, with the regression line confirming the protective effect. The bar chart (right) displays odds ratios from the logistic model, with private inpatient care dominating.

5.5 RQ7: LASSO Variable Selection

LASSO regression with 5-fold cross-validation selected an optimal regularisation parameter of $\alpha = 0.011$ and retained **31 out of 51 candidate features**, setting the remaining 20 coefficients exactly to zero. This confirms that many demographic and socioeconomic variables provide redundant information once the key drivers are accounted for.

Table 6: RQ7: Top 10 LASSO Predictors

Feature	Coef.	Direction
state_region_A	1.226	↑ Higher OOP
state_region_C	−0.972	↓ Lower OOP
religion_Sikh	−0.940	↓ Lower OOP
had_ip_visit	0.880	↑ Higher OOP
op_private	0.735	↑ Higher OOP
ip_private	0.651	↑ Higher OOP
uhc_tercile_Low	0.594	↑ Higher OOP
uhc_index	0.526	↑ Higher OOP
religion_infreq.	0.441	↑ Higher OOP
total_hh_spend	−0.416	↓ Lower OOP
31 of 51 features retained; α=0.011		

5.5.1 Interpretation

The LASSO results reveal a clear hierarchy of predictor importance. **Geographic region** emerges as the single strongest predictor—state region A (Northern states) has the largest positive coefficient (1.226), while region C has a strong negative coefficient (−0.972). This reflects enormous inter-state variation in healthcare costs, infrastructure quality, and pricing practices across India.

Healthcare utilisation variables form the second most important cluster: inpatient visits (0.880), private outpatient use (0.735), and private inpatient use (0.651) are all retained with large positive coefficients. This aligns with the OLS findings from RQ1–RQ3.

District-level infrastructure (UHC tercile and index) also survives LASSO selection, reinforcing the importance of place-based health system quality. Notably, total household spending has a *negative* coefficient (−0.416), meaning wealthier households have lower OOP after accounting for utilisation patterns—consistent with their greater ability to access insurance and subsidised care.

Demographic variables such as age, gender, and occupation are either eliminated or retained with small coefficients, suggesting they contribute relatively little explanatory power once facility type and geography are controlled.

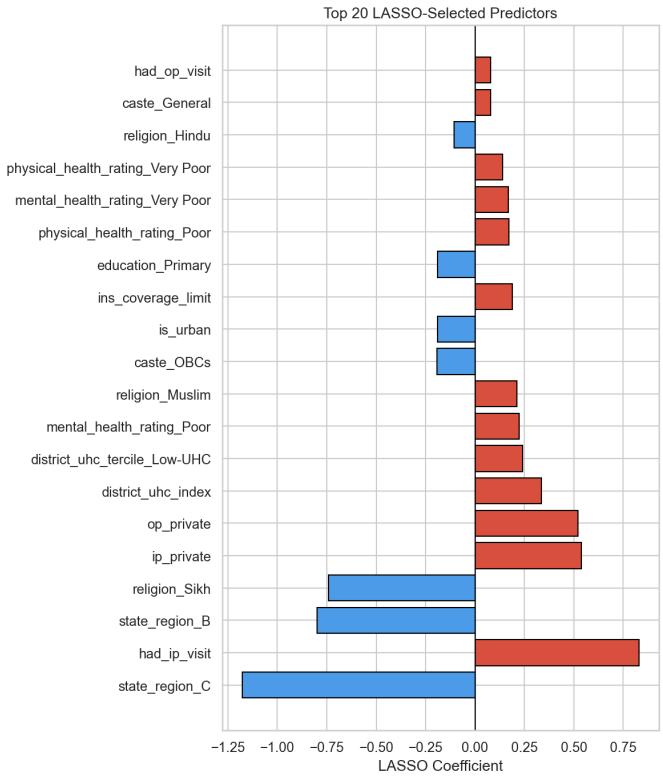

Figure 5: RQ7: Top 20 LASSO-selected predictors ranked by coefficient magnitude. Red bars indicate features that increase OOP; blue bars indicate protective features. Geographic and facility-type variables dominate.

5.6 RQ8: Outpatient vs Inpatient Gap

Table 7: RQ8: OLS — IP–OP Gap Determinants

Variable	Coef.	SE	p
ip_private	21640	1779	0.000
uhc_index	105	45	0.019
is_urban	−1470	686	0.032
is_male	−1393	787	0.077
is_insured	829	1163	0.476

N=17,206; R^2=0.037; HC3 SEs

5.6.1 Interpretation

The IP–OP gap measures the difference between inpatient and outpatient expenditure for each household. A positive gap means inpatient costs substantially exceed outpatient costs.

Private inpatient care is the dominant predictor, adding INR 21,640 to the gap ($p < 0.001$). This means that private hospitalisation costs roughly INR 21,640 more than would be expected from outpatient spending patterns alone, reflecting the high cost of surgery, intensive care, and extended hospital stays in private facilities.

Urban residence narrows the gap by INR 1,470 ($p = 0.032$), suggesting that urban households have access to more affordable outpatient alternatives (e.g., polyclinics, diagnostic centres) that partially substitute for expensive inpatient care.

Insurance is **not significant** ($p = 0.476$), indicating that current insurance schemes do not meaningfully reduce the IP–OP expenditure differential. This is concerning because inpatient costs are precisely where catastrophic spending is most likely.

The low R^2 (0.037) indicates that the gap is driven largely by idiosyncratic factors—individual illness severity, specific treatments received, and hospital choice—that are not captured by the available demographic variables.

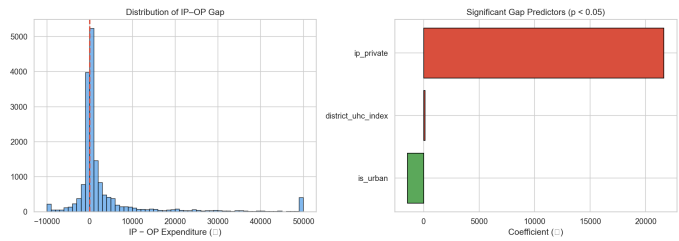

Figure 6: RQ8: The left histogram shows the distribution of the IP–OP expenditure gap, which is heavily right-skewed with a long tail of extreme inpatient costs. The right panel identifies statistically significant predictors, with private inpatient care dominating.

6 Discussion

6.1 Insurance Effectiveness

Our analysis reveals a nuanced and somewhat contradictory picture of insurance in India. The OLS model (RQ1) shows

a significant protective effect: insurance reduces OOP by 54% after controlling for observed confounders. However, the logistic models (RQ4–RQ6) consistently show that insured individuals are *more* likely to face catastrophic expenditure (OR = 1.22–1.50).

This **insurance paradox** is consistent with two well-documented mechanisms in health economics literature:

- **Adverse selection:** Individuals with higher expected healthcare needs disproportionately enrol in insurance schemes, particularly voluntary ones. Since insurance does not fully cover costs, these individuals still face high absolute expenditure.
- **Moral hazard:** Insurance reduces the marginal cost of care-seeking, encouraging utilisation of more expensive services—particularly private hospitals—that may not have been accessed without coverage.

The significant insurance × urban interaction (RQ2) further suggests that insurance benefits are geographically uneven, with urban areas offering a weaker protective effect, likely due to higher private facility costs that exceed reimbursement limits.

6.2 Public–Private Divide

The magnitude of private facility effects is striking and consistent across all models. Private outpatient and inpatient care increase OOP by 500–570% (RQ3), and private inpatient usage has an OR of 89 for catastrophic expenditure (RQ6). These findings underscore the urgent need for cost regulation in private healthcare.

India's Clinical Establishments Act (2010) provides a framework for price regulation, but implementation has been limited to a few states. Our results suggest that expanding price transparency and capping charges for common procedures could substantially reduce catastrophic expenditure. The LASSO analysis (RQ7) further confirms that facility type is among the top three predictors of OOP, making it a high-impact target for policy intervention.

6.3 Geographic and Structural Factors

District UHC infrastructure consistently emerges as a protective factor across fixed-effects models ($p < 0.001$ in RQ1, RQ4, RQ6). However, the mixed-effects model (RQ9) reveals that this effect is largely absorbed by district random intercepts, suggesting that UHC index is a proxy for broader, unobserved district-level characteristics such as governance quality, local economic conditions, and healthcare supply-side factors.

The LASSO results further highlight the importance of geography: state region is the single strongest predictor, with Northern states (region A) showing substantially higher OOP. This inter-state variation reflects India's decentralised health system, where states have significant autonomy in healthcare delivery and financing.

The random intercept variance (0.033) confirms that approximately 3.3% of total variation in catastrophic expenditure is attributable to district-level factors, justifying the use of hierarchical models and suggesting that place-based interventions could be effective.

6.4 Limitations

- **Cross-sectional design:** The data represent a single point in time, precluding causal inference. The insurance effects, in particular, may be confounded by unobserved health status.
- **Multicollinearity:** The condition number of 2,350 in the RQ1 model indicates moderate multicollinearity among predictors. While HC3 robust standard errors mitigate bias in inference, individual coefficient estimates may be unstable.
- **Low explanatory power in gap model:** The R^2 of 0.037 in RQ8 suggests that most variation in the IP–OP gap is driven by unobserved factors (illness severity, specific treatments, hospital choice).
- **Measurement error:** Self-reported expenditure data may contain recall bias, particularly for outpatient costs that are typically smaller and more frequent.
- **Consumption proxy:** Using food + education spending as an income proxy may underestimate true household resources, potentially inflating catastrophic expenditure rates.

7 Conclusion

This study provides comprehensive evidence on the determinants of out-of-pocket healthcare expenditure in India, using multiple complementary statistical methods on a nationally representative dataset of 40,070 respondents. Three overarching findings emerge:

1. **Insurance provides partial but insufficient protection.** While insurance reduces average OOP by 54%, it does not prevent catastrophic expenditure—and insured individuals are actually *more* likely to face catastrophic spending due to adverse selection. Current coverage limits and reimbursement rates appear inadequate for serious illnesses requiring hospitalisation.
2. **Private healthcare is the dominant cost driver.** Private facility effects (500–570% increase in OOP; OR = 89 for catastrophic expenditure) are an order of magnitude larger than any other predictor. Without regulatory intervention on private pricing, no insurance scheme can sustainably reduce OOP burden.
3. **Geography and infrastructure matter.** District-level UHC infrastructure, state region, and local

healthcare supply are among the strongest predictors of expenditure. The significant between-district variance in the mixed-effects model confirms that place-based investments in public healthcare can meaningfully reduce financial burden.

7.1 Policy Recommendations

Based on our findings, we propose the following interventions:

- **Strengthen public healthcare infrastructure:** Expanding PHCs, CHCs, and district hospitals would provide affordable alternatives to private facilities, directly targeting the largest cost driver.
- **Regulate private facility pricing:** Implementing transparent pricing norms and capping charges for common procedures under the Clinical Establishments Act.
- **Expand insurance coverage limits:** Current Ayushman Bharat coverage of INR 5 lakh per family per year may be insufficient for complex hospitalisations. Increasing limits and reducing copayment requirements would strengthen the protective effect.
- **Target low-UHC districts:** Directing health system investments to districts with low UHC index scores, where catastrophic expenditure rates are highest.

7.2 Future Work

Future studies should leverage panel data to establish causal effects of insurance and employ instrumental variable approaches to address endogeneity. Decomposing the district random effects into measurable infrastructure components would also help identify specific investment priorities.

Appendix: Software and Reproducibility

All analyses were conducted using Python 3.14 with the following packages: `statsmodels` (OLS, logistic, mixed-effects), `scikit-learn` (LASSO), `scipy` (Mann–Whitney U test), `pandas`, `numpy`, `matplotlib`, and `seaborn`. Heteroscedasticity-robust standard errors (HC3) were used for all OLS models. The analysis script (`run_analysis.py`) and all output files are included in the `oop_outputs/` directory.

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